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Successful Conversion of a Drill Cuttings Reinjection (CRI) Well into a Produced Water Disposal (PWD) Well: A Unique Case Study from the Middle East

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Abstract

To address the high demand for produced water management, a Cuttings Reinjection (CRI) well (well-1) was converted into a Produced Water Disposal (PWD) well. The study outlines the methodology, results, challenges, and cost savings associated with this conversion.

The target reservoir for existing PWD wells consists of multiple layers of limestone and dolomite. An extensive study was conducted to plan the PWD wells in this target zone, supported by production logging tools (PLT) & core data. Initial injectivity results indicated that well-1's performance (bare footed completion) was fair, and it was comparable to nearby well's performance, having same completion design. The injectivity test was also compared with an existing CRI well (well-2) in the same area, having different completion design. After the initial injectivity test, well-1 remained active as a CRI for five years with low injection rates, followed by another five years of being shut-in due to surface facility constraints. Given the high demand for produced water management, it was considered whether to evaluate a new PWD well locations or opt to a cost-effective option to reactivate well-1 as PWD, given the surface slots are limited. Because of this, an integrated approach was used for evaluation, focusing on geological (such as Temp logs, etc.) and geophysical data (such as seismic, etc.). Injectivity tests were performed in well-1 at different depths, guided by attribute maps. Due to poor performance results, the injectivity test was repeated after each acid stimulation job till the desired results were achieved.

Following a detailed geological and geophysical evaluation, two prolific zones were identified as potential candidates for sustainable injection. Seismic reservoir characterization revealed the distribution of cavities (karst collapse features) within the target zones, which correlated well with dynamic mud losses. Based on the performance of nearby wells and results from seismically driven attributes, optimum locations were considered for new PWD wells. The attribute map also showed that well-1 targets a highly porous zone, as confirmed by the initial injectivity results. However, the first injectivity test conducted after ten years yielded extremely poor results. The injectivity test after third acid stimulation job showed slight uplift in improvement, though still lower than the initial injectivity performance. Subsequently injectivity tests were repeated after every acid stimulation job. The last injectivity test after the seventh acid job showed a two-fold improvement in water injection compared to the initial results. Based on the strong injectivity

result after the seventh acid stimulation job, it was decided to activate well-1 as a PWD. Well-1 has now been active as a PWD for over a year, showing sustainable performance. This conversion resulted in saving a surface slot along with over \$10MM incremental savings in CAPEX/OPEX.

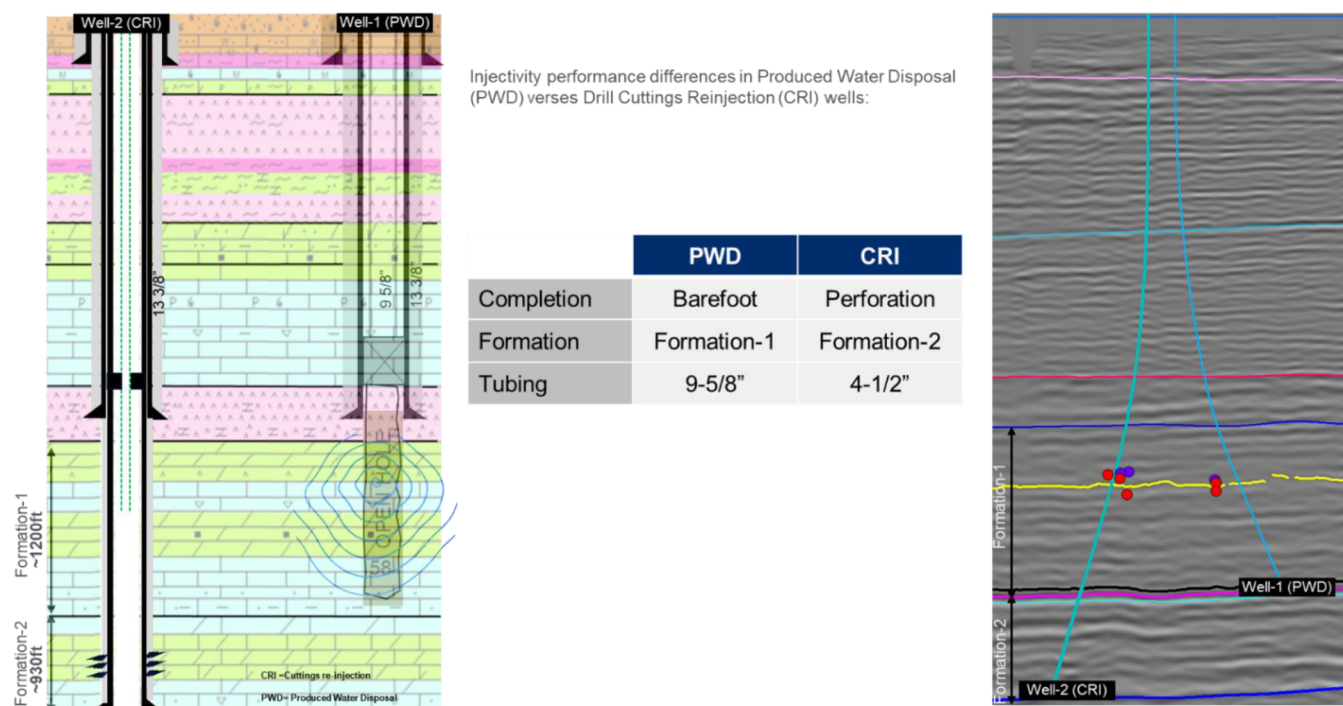
Converting a CRI into a PWD well is not common due to completion design constraints, injection performance, and target Formation heterogeneity. Consequently, this unique study focused on post-conversion results to validate that the well can be switched from a CRI to a PWD or vice-versa without any changes in completion design and extra cost implications.

Introduction

The study area covers a vast offshore oil field in the Middle East. A significant source of hydrocarbon reserves in this region comes from carbonate platforms, ranging from Upper Permian to Oligocene reservoirs (Ahmad et al., 2022). Water injection has been globally utilized for pressure maintenance to enhance hydrocarbon recovery and to dispose of unwanted produced water from oil and gas wells in an environmentally responsible manner (Bennion et al., 2000). Carbonate rocks with karst collapse features have shown high injectivity for produced water disposal. These rocks require minimal water quality screening before injection, as the risk of reservoir damage or permeability reduction is low.

The field's current water production is moderate to high, with an expected increase to nearly double its current rate in the next decade (Ahmad et al., 2024). This increase necessitates substantial investment in water management facilities, additional water disposal wells, and most importantly, investigating subsurface reservoirs capable of accommodating large amounts of water. Effective management of produced water is a global concern that requires advanced surface and subsurface investigations. When reinjecting into the parent Formation is not feasible, it is important to consider an overburden Formation capable of accommodating large amounts of produced water without harming the environment or damaging the reservoir. The Formation damage arising from the injection of produced water containing impurities is usually a significant concern that poses challenges for maintaining good injectivity of disposal wells.

Based on detailed geological and geophysical investigations, six disposal wells have been drilled in the study area, four of which are PWD wells and two are CRI wells. Almost all of them observed dynamic mud losses while drilling into Formation-1. Formation-1 has been evaluated in detail and divided into four broad zones (unit-1 to unit-4). It is observed that most of the drilling challenges (e.g., dynamic mud losses) begin to occur in conformable unit-2, which is predominantly calcite (limestone) with dolomite intercalations. PLTs (available for Well-1 only) also confirmed high injectivity in unit-2 (Ahmad et al., 2024). The effective zones within Formation-1 have been identified through extensive geological studies, including Formation evaluation, dynamic data, seismic interpretation, and attribute analysis (Ahmad et al., 2022). Produced water is currently disposed of in Formation-1, and cuttings and other waste are being disposed of in Formation-2. Karst collapse features are found to be the most suitable for good injectivity (Ahmad et al., 2023). PWD and CRI wells have different configurations in terms of tubing size, target Formation, and lower completion (Fig. 1).



Well-1 was drilled as a CRI well with the same completion design as that used for PWD wells in Formation-1. The well observed dynamic mud losses while drilling, suggesting that it is targeted at the right place (karst features) in Formation-1. An initial injectivity test was performed that showed good results (Fig. 2). After the initial injectivity test, Well-1 remained active as a CRI for five years with low injection rates (400-600 bpd), followed by another five years of being shut-in due to surface facility constraints. Given the high demand for produced water management (Ahmad et al, 2022), it was considered whether to evaluate new PWD well locations or opt for a cost-effective option to reactivate Well-1 as PWD, given the limited surface slots. The objective was to reduce capital expenditure by improving water disposal with fewer wells, less reservoir damage, and minimal environmental concerns. In this regard, a detailed subsurface investigation was carried out with a main emphasis on well performance and other dynamic information from nearby wells. It was also expected that reviving the well after ten years might not perform well due to cuttings/waste that might have adversely affected the near borehole. An injectivity test performed after ten years showed very poor outcomes. A decision was made to perform acid stimulation and repeat the injectivity test. There was negligible improvement in the second injectivity test. Acid stimulation was repeated three times, and subsequent injectivity tests were also performed until the injectivity performance reached the initial injectivity. Though injectivity was matching with the initial injectivity, it was less compared to the other nearby wells. Therefore, the acid stimulation job was repeated seven times, and subsequent injectivity tests showed remarkable improvement in injectivity, comparable with nearby wells. The well was then put into injection based on the available quota, without any issues observed.

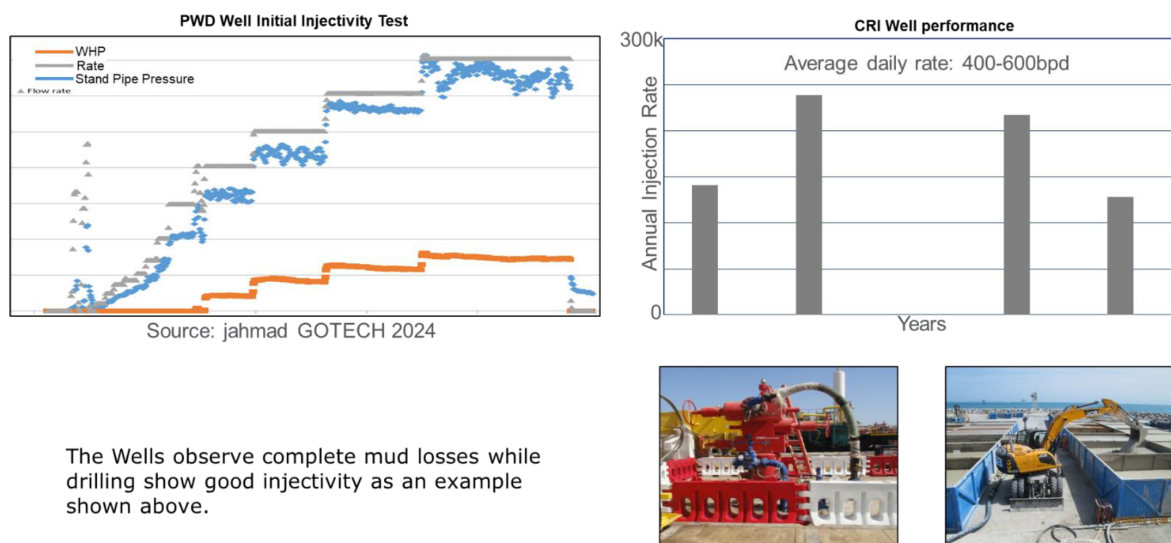


Figure 2—The initial injectivity test results are shown on the left. The performance history of the well during the CRI phase is displayed on the right along with waste storage and CRI units.

Well Performance

The initial injectivity test for Well-1 was conducted shortly after drilling and completion, using sea water. The pumping rate increased from 3kbd to 50kbd, with WHP changes ranging from 50psi to 1360psi. The results confirm that Well-1 can sustain long-term injection in Formation-1 (Fig. 2). Originally a test CRI well with large tubing (9-5/8"), it was soon utilized for injecting cuttings, WBM (Water Based Mud), and other waste. Fig. 2 shows the waste storage and cuttings reinjection units. The first year's injection rates were low due to a single rig but nearly doubled in the second year, averaging 182,500 bbls annually (500BPD). After five years, waste management shifted to Well-2, targeting Formation-2, leaving Well-1 inactive for approximately five years.

The current water production rate of the field is moderate to high, with a projected increase expected to nearly double in the next decade (Ahmad et al., 2024). This anticipated growth necessitates significant investment in water management infrastructure, additional water disposal wells, and critical examination of subsurface reservoirs capable of accommodating substantial volumes of water. To address the high demand for produced water management, an evaluation was conducted to determine whether new PWD well locations should be assessed or if reactivating Well-1 as PWD was a cost-effective alternative, given the limited surface slots available. An integrated approach was employed, focusing on geological and geophysical data.

Following five years of CRI injection and subsequently remaining dormant for the next five years, injectivity tests were performed at Well-1 at varying depths, guided by the attribute maps within four units in Formation-1. The first injectivity test indicated poor results, represented by the pink line in Fig. 3. This outcome was likely due to blockage of pore spaces around the borehole after years of CRI injection. Repeating the injectivity test after an acid stimulation job yielded similar results to the first test.

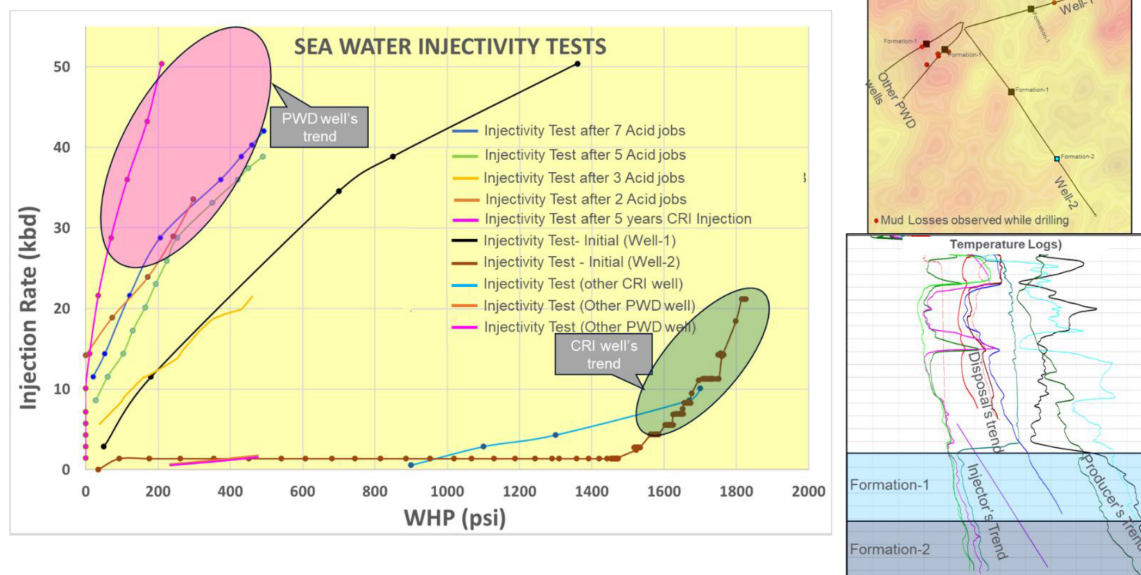


Figure 3—On the left, a comparison of sea water injectivity tests for all nearby wells is presented, along with the injectivity performance of Well-1 following various acid stimulation jobs. RMS amplitude map is shown for unit-2, including well trajectories (black) and dynamic mud losses (red dots). TEMP logs for nearby injectors and producers are also provided to illustrate various trends.

However, improvements were noted following the third acid stimulation job, with injectivity levels approaching those observed a decade ago. Subsequent acid stimulation jobs were repeated for a fifth and seventh time, each followed by injectivity tests, which demonstrated significant enhancement in injectivity. This improvement rendered Well-1's performance comparable to other PWD wells targeting karst collapse features in Formation-1, where complete mud losses were encountered during drilling. The RMS amplitude map demonstrates the distribution of karst collapse features, characterized by bull's eye patterns, within Formation-1 (Unit-2). Notably, Well-1 intersects the boundary of these karst features, with the depth of the mud losses indicated by the red points.

Furthermore, the plot (Fig. 3) shows injectivity test results of existing CRI wells (red and cyan lines) targeting Formation-2. Given the smaller tubing size (4-1/2") and deeper perforation interval in Formation-2, relatively favorable injection rates were observed at over 1500 psi WHP. A notable difference in injectivity trends exists between PWD and CRI wells. Temperature logs run on adjacent wells revealed three distinct trends: the injector's low gradient trend, the producer's high gradient trend, and a median trend for disposal wells. These findings also highlight certain integrity-related concerns but that is out of scope for this paper.

Well-1's performance after reactivation as PWD has been compared with nearby PWD wells (Fig. 4). During the CRI injection phase, rates were low (400-600 BPD). No injection periods are marked in grey. During the PWD phase, Well-1's performance (cyan line) and the other wells' injection histories are shown. Since the water produced was split among three wells, Well-1 was not tested to its full capacity but maintained consistent performance throughout the injection period.

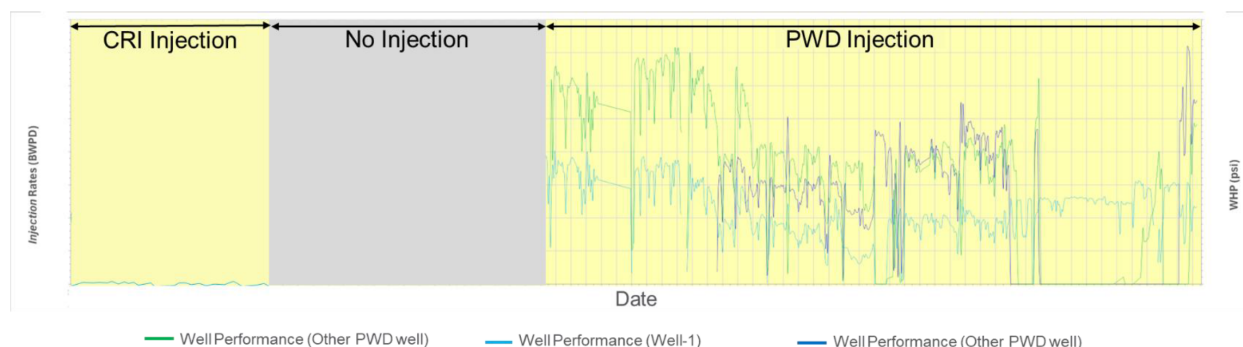


Figure 4—The plot shows the various phases of injection in Well-1: five years of low-rate CRI injection, five years of no injection, followed by PWD injection. It also includes a comparison with the performance of a nearby well during PWD injection phase.

Discussion

In the study area, converting a CRI well into a PWD well had not been previously tested due to three primary limitations:

- Change in the lower completion design (barefoot for PWD wells versus perforations for CRI wells)
- Differences in Formations (Formation-1 for PWD wells versus Formation-2 for CRI wells)
- Tubing size (9-5/8" for PWD wells versus 4-1/2" for CRI wells)

The study area currently has four PWD and two CRI wells with different configurations. The unique situation with Well-1 was that its completion design was barefoot and tubing size was 9-5/8", although it targeted Formation-1 as a CRI well. After five years of low injection rates followed by a shut-in period, reactivating Well-1 was considered to manage produced water. An initial injectivity test showed poor results, leading to multiple acid stimulations until satisfactory injectivity was achieved. Subsequent tests showed improved injectivity comparable to nearby wells, allowing Well-1 to resume injection without issues due to larger tubing and barefoot completion.

Additionally, this case study highlights the optimization potential of intermittently converting a CRI well into a PWD well or vice versa as needed. This can result in significant cost savings in terms of drilling additional wells and surface slots.

Conclusions

The study area covers a major offshore oil field in the Middle East. Hydrocarbon reserves primarily come from carbonate platforms dating from Upper Permian to Oligocene reservoirs. The current water production is expected to double in the next decade, requiring significant investment in water management facilities and disposal wells. Effective produced water management requires advanced investigations for suitable overburden Formations.

Six disposal wells have been drilled: four PWD and two CRI wells. Most drilling challenges occur in unit-2 of Formation-1, which consists mainly of limestone with dolomite intercalations. PLTs confirmed high injectivity in unit-2. Produced water is disposed of in Formation-1, with cuttings and waste in Formation-2. Extensive geological studies identified effective zones within Formation-1, including karst collapse features.

Well-1, initially drilled as a CRI well, faced dynamic mud losses indicating correct targeting in unit-2 within Formation-1. An initial injectivity step rate test was conducted on Well-1 soon after its drilling and completion. The test, utilizing seawater, involved gradually increasing the pumping rate from 3kbd to 50kbd, with a corresponding WHP change ranging from 50psi to 1360psi. Results confirmed Well-1's

capability for long-term injection into Formation-1. Originally planned as a test CRI well with large tubing size (9-5/8"), Well-1 began injection operations shortly after the test, handling cuttings, WBM, and other waste. Annual injection rates were initially low due to the operation of only one rig, but they almost doubled in the subsequent year, averaging 182,500 barrels per annum (500 BPD).

After five years of CRI injection in Well-1, all associated cuttings and waste management were redirected to Well-2, targeting deeper Formation-2. Consequently, Well-1 remained inactive for approximately five years.

The field currently exhibits moderate to high water production, with projections indicating a potential doubling of this rate within the next decade. This expected increase necessitates significant investment in water management infrastructure, including additional water disposal wells and the exploration of subsurface reservoirs capable of accommodating substantial water volumes. Given the high demand for produced water management, consideration was given to either evaluating new PWD well locations or reactivating Well-1 as a more cost-effective option due to limited surface slots. An integrated approach, focusing on geological and geophysical data, was adopted for this evaluation.

Post-reactivation performance of Well-1 as a PWD has been plotted and compared with nearby PWD wells. During the CRI injection phase, the injection rates were low (400-600bpd) with non-injection periods marked in grey. In the PWD injection phase, Well-1's performance was consistent despite receiving a lower water allocation compared to other PWD wells, owing to the division of produced water among three wells.

Well-1 has now been active as a PWD for over a year, showing sustainable performance. This conversion resulted in saving a surface slot along with over \$10MM incremental savings in CAPEX/OPEX.

Acknowledgement

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Nomenclature

BPD	= Barrels Per Day
CRI	= Cuttings Re-Injection
PLT	= Production Logging Tools
PSI	= Pounds per inch
PWD	= Produced Water Disposal
WBM	= Water Based Mud

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